

Case Study: Ticket SLA Analytics Dashboard for National Skill Development Corporation

Project Overview

Developed an advanced Ticket SLA Monitoring and Analytics Solution using Zoho Analytics integrated with Zoho Desk for the Skill India Digital Hub support operations.

The project focused on building an automated SLA calculation engine from raw ticket status history and creating management dashboards for operational monitoring.

Business Challenge

The client faced multiple challenges:

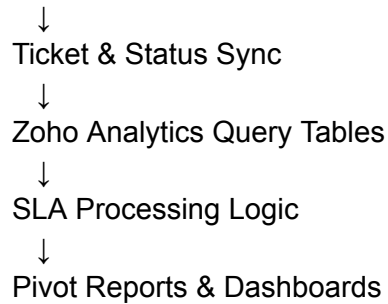
- No centralized SLA reporting
- Manual SLA calculations
- Difficulty tracking response timelines
- Complex ticket lifecycle
- Multiple status transitions
- No visibility into department-wise and agent-wise SLA performance
- Inability to measure support efficiency accurately

Since Zoho Desk stores ticket updates as event history logs, calculating actual SLA required custom SQL transformation logic inside Zoho Analytics.

Solution Architecture

Data Flow

Zoho Desk



Step 1 — Extract Ticket Status Timeline

The first step was to combine:

- Ticket master data
- Status history
- Department details

to create a complete ticket activity timeline.

Query 1 — Ticket Status Timeline

```
SELECT
    Status."Ticket" as 'Ticket',
    Tickets."Created Time" as 'Created Time',
    Status."Status Name" as 'Status Name',
    Status."Status updated time" AS 'Status updated time'
FROM "Status" AS 'Status'
JOIN "Tickets" AS 'Tickets'
    ON Status."Ticket" = Tickets."ID"
LEFT JOIN "Department"
    ON "Department"."ID" = "Tickets"."Department"
WHERE "Department"."Department Name" = 'Skill India Digital Hub'
```

Purpose of Query

This query:

- Extracted ticket lifecycle events

- Mapped status history with ticket creation time
 - Filtered department-specific tickets
 - Created the base dataset for SLA calculations
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Step 2 — Create Business Status Classification

The next challenge was identifying meaningful SLA events.

We classified ticket statuses into business categories:

Ticket Status	Status Type
Open	OPEN
Customer Responded	RESPONSE
Escalated	RESPONSE
Pending for Approval	RESPONSE
Pending on User	RESPONSE
Resolved	RESPONSE
Others	OTHER

Step 3 — Remove Duplicate Status Transitions

Tickets often had repeated status updates like:

Open → Open → Open → Customer Responded

This could incorrectly increase SLA calculations.

To solve this:

- Window Functions
- **LAG()** Logic
- Status comparison

were implemented.

Query 2 — Prepare Ticket Status Data

```
SELECT
  t."Ticket" as 'Ticket',
  t.Status_Type as 'Status_Type',
  t."Status updated time" AS 'Status updated time'
FROM (
  SELECT
    "Ticket",

    CASE
      WHEN "Status updated time" = "Created Time"
        THEN 'OPEN'

      WHEN "Status Name" = 'Open'
        THEN 'OPEN'

      WHEN "Status Name" IN
        (
          'Customer Responded',
          'Escalated',
          'Pending for Approval',
          'Pending on User',
          'Resolved'
        )
        THEN 'RESPONSE'

      ELSE 'OTHER'
    END AS Status_Type,
```

```

"Status updated time",

LAG(
  CASE
    WHEN "Status updated time" = "Created Time"
      THEN 'OPEN'

    WHEN "Status Name" = 'Open'
      THEN 'OPEN'

    WHEN "Status Name" IN
      (
        'Customer Responded',
        'Escalated',
        'Pending for Approval',
        'Pending on User',
        'Resolved'
      )
      THEN 'RESPONSE'

    ELSE 'OTHER'
  END
) OVER(
  PARTITION BY "Ticket"
  ORDER BY "Status updated time"
) AS Prev_Status

FROM "Ticket_status"
) t

WHERE t.Status_Type <> t.Prev_Status
OR t.Prev_Status IS NULL

ORDER BY
  t."Ticket",
  t."Status updated time"

```

What This Query Solved

Key Logic

LAG() OVER(PARTITION BY Ticket ORDER BY Status updated time)

This helped:

- Detect previous ticket status
 - Remove duplicate status transitions
 - Identify valid SLA movement
 - Clean ticket lifecycle data
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Step 4 — Calculate SLA Response Time

After preparing OPEN and RESPONSE events:

- OPEN timestamp was matched with the next RESPONSE timestamp
 - Difference between timestamps became SLA time
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Query 3 — Calculate Response SLA

```
SELECT
  o."Ticket" as 'Ticket',
  o."Status updated time" AS "Open Time",
  MIN(r."Status updated time") AS "Response Time"

FROM "prepared_ticket_data" o

JOIN "prepared_ticket_data" r
  ON o."Ticket" = r."Ticket"
  AND r."Status_Type" = 'RESPONSE'
  AND r."Status updated time" > o."Status updated time"

WHERE o."Status_Type" = 'OPEN'

GROUP BY
  o."Ticket",
  o."Status updated time"
```

ORDER BY
o."Ticket",
o."Status updated time"

SLA Formula

SLA\ Hours = Response\ Time - Open\ Time

Output Generated

This query generated:

- Open Time
 - First Response Time
 - SLA Duration
 - Ticket-wise SLA tracking
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Step 5 — Monthly SLA Aggregation

The calculated SLA values were aggregated month-wise for management reporting.

Query 4 — Month Wise SLA Report

```
SELECT  
  ROW_NUMBER() OVER(ORDER BY t."S.No") AS "S.No",  
  t."Year",  
  t."Month No",  
  t."Month" as 'Month',  
  t."Average SLA Time" as 'Average SLA Time'
```

FROM (

```

SELECT
    (YEAR("Created Time") * 100 + MONTH("Created Time")) AS "S.No",

    YEAR("Created Time") AS "Year",

    MONTH("Created Time") AS "Month No",

    CONCAT(
        MONTHNAME("Created Time"),
        '(',
        YEAR("Created Time"),
        ')'
    ) AS "Month",

    CONCAT(
        FLOOR(AVG("SLA (Hours)") / 24),
        ' Days ',

        FLOOR(MOD(AVG("SLA (Hours)"), 24)),
        ' Hours ',

        FLOOR(
            (
                AVG("SLA (Hours)")
                - FLOOR(AVG("SLA (Hours)"))
            ) * 60
        ),
        ' Minutes'
    ) AS "Average SLA Time"

FROM "Ticket Wise Total SLA"

GROUP BY
    (YEAR("Created Time") * 100 + MONTH("Created Time")),
    YEAR("Created Time"),
    MONTH("Created Time"),
    CONCAT(
        MONTHNAME("Created Time"),
        '(',
        YEAR("Created Time"),
        ')'
    )
) AS t

```

Average SLA Formula

Average\ SLA = $\frac{\text{sum SLA\ Hours}}{\text{Total\ Tickets}}$

Dashboard Features

Executive KPI Dashboard

- Total Tickets
 - Open Tickets
 - Closed Tickets
 - Incident Tickets
 - Escalated Tickets
 - SLA Compliance %
 - Average Response Time
 - Average Resolution Time
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Reports Developed

SLA Reports

- Month-wise SLA Trend
- SLA Breach Analysis
- Response Time Monitoring
- Ticket Aging Analysis

Operational Reports

- Department-wise Distribution
- Status-wise Ticket Analysis
- Priority-wise Reports
- Escalation Tracking

Agent-Wise SLA Enhancement

After successful deployment of overall SLA reporting, the system was extended for Agent Performance Analytics.

Additional Enhancements

- Agent-wise SLA Calculation
- Ticket Handling Time
- Average Resolution Time per Agent
- Agent Productivity Dashboard
- SLA Breach by Agent
- Escalation Ratio

This enabled management to:

- Measure individual performance
- Optimize support workload
- Identify bottlenecks
- Improve support efficiency

Final Reporting Layer

The processed data was finally used in:

- Pivot Tables
- Interactive Dashboards
- KPI Widgets
- Drill-down Reports
- Scheduled MIS Reports

inside Zoho Analytics.

Technologies Used

- Zoho Analytics
 - Zoho Desk
 - SQL Query Tables
 - Window Functions
 - Pivot Analytics
 - Dashboard Automation
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Business Outcome

Key Achievements

- Automated SLA reporting system
 - Accurate response time calculation
 - Real-time ticket monitoring
 - Reduced manual reporting effort
 - Improved operational transparency
 - Enhanced agent accountability
 - Better decision-making through analytics
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My Role in the Project

Responsibilities

- Requirement gathering
- SLA logic design
- SQL query development
- Zoho Analytics dashboard development
- BRD preparation
- Client communication
- Stakeholder management
- Project coordination
- Report optimization
- User training & demonstrations

POC :

Month Wise Ticket SLA Report

Displays the average SLA response time for tickets on a monthly basis, helping track performance trends and identify improvements in response efficiency over time.

S.No	Month(Year)	Average SLA Time
1	April (2025)	1 Days 1 Hours 49 Minutes
2	May (2025)	1 Days 1 Hours 22 Minutes
3	June (2025)	1 Days 0 Hours 10 Minutes
4	July (2025)	0 Days 22 Hours 22 Minutes
5	August (2025)	1 Days 2 Hours 16 Minutes
6	September (2025)	1 Days 0 Hours 35 Minutes
7	October (2025)	1 Days 1 Hours 12 Minutes
8	November (2025)	1 Days 3 Hours 32 Minutes
9	December (2025)	0 Days 22 Hours 57 Minutes
10	January (2026)	0 Days 22 Hours 14 Minutes
11	February (2026)	0 Days 20 Hours 28 Minutes
12	March (2026)	0 Days 15 Hours 40 Minutes
13	April (2026)	0 Days 19 Hours 17 Minutes
14	May (2026)	0 Days 13 Hours 5 Minutes